MIL-P-23281(CG)
4 May 1962
SUPERSEDING
CG Spec. CGS-52P-4c
15 April 1958

MILITARY SPECIFICATION

PRIMER, VINYL-RED LEAD (FOR BRUSH OR SPRAY)

1. SCOPE

1.1 Scope.- This specification covers vinyl-red lead primer for use by brush or spray over pretreatment coating (Formula No. 117, Specification MIL-P-15328).

1.2 Classification.- The primer shall be furnished in the following colors, as stipulated:

Orange
Brown

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standard, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-M-268	Methyl Isobutyl Ketone (For use in Organic Coatings)
TT-P-143	Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of
TT-P-350	Pigment, Lampblack-Dry
TT-R-191	Red Lead, Dry and Paste-in-Oil
TT-T-548	Toluene; Technical
TT-T-656	Tricresyl Phosphate

FSC 8010

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MILITARY

MIL-P-15328 Primer, Pretreatment (Formula No. 117 for Metals)

MIL-T-19588 Toluene-Methyl Isobutyl Ketone Mixture

STANDARDS

FEDERAL

FED-STD-141 Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The primer shall consist of ingredients conforming to the applicable specifications in the proportions shown in Table I. The formula in Table I is given to approximately 100 gallons to allow for normal manufacturing loss.

Table I - Formula for Vinyl-Red Lead Primer

Ingredients	Pounds per 100 gallons
Red lead (type I, grade 97 of Spec. TT-R-191)[1]	220
Vinyl resin[2]	145
Tricresyl phosphate (Spec. TT-T-656)	15
Methyl isobutyl ketone (Spec. TT-M-268)	295
Toluene (Spec. TT-T-548)	295

[1]For brown color — red lead and lampblack (see 3.4.2).

[2]The resin shall be a hydroxyl containing vinyl chloride-acetate copolymer. It shall contain 89.5 to 91.5 percent vinyl chloride, 5.3 to 7.0 percent vinyl alcohol, and 2.0 to 5.5 percent vinyl acetate. The resin shall have a specific gravity of 1.35 minimum. Material shall be furnished as a powdered solid not less than 98 percent of which shall pass through a No. 20 (840 micron) U. S. Standard Sieve. An 18 percent solution of the resin in methyl isobutyl ketone shall be no darker than 1.0 gram of potassium dichromate dissolved in one liter of water.

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3.2 Manufacture.- The component raw materials shall be mixed and ground as required to produce a product which is uniform, homogeneous, free from grit, entirely suitable for the purpose intended, and in full conformity to the requirements of this specification.

3.3 Quantitative requirements.- The primer shall meet the quantitative requirements shown in Table II.

Table II - Quantitative Requirements

Characteristics	Requirements	
	Minimum	Maximum
Pigment, percent by weight	21.0	24.0
Volatiles, percent by weight	59.0	63.0
Nonvolatile vehicle, percent by weight (calculated by difference)	15.0	18.0
Water, percent by weight	---	0.5
Coarse particles and skins (as residue retained on standard No. 325 (44 micron-U. S. Standard Sieve Series sieve), percent by weight of paint	---	0.5
Viscosity, krebs units	63	78
Weight per gallon, pounds	9.2	9.8
Fineness of grind	5	---
Flash point (Pensky-Martens Closed Cup)	40°F	---
Time of setting to touch, minutes	---	15
Time of drying hard, minutes	---	30
Red lead (Pb_3O_4), percent by weight of pigment	97	---

3.4 Qualitative requirements.- The primer shall meet the following qualitative requirements:

3.4.1 Odor.- The odor shall be normal for the volatiles permitted.

3.4.2 Color.-

3.4.2.1 Orange.- The orange primer shall be the characteristic color of red lead conforming to Type I, Grade 97 of Specification TT-R-191.

3.4.2.2 Brown.- The brown primer shall be a differentiating brown color obtained by the addition of .25 percent (of the total pigment) of lampblack conforming to Specification TT-P-350.

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3.4.3 Compatibility.- There shall be no evidence of incompatibility of any of the ingredients of the primer when tested as specified in 4.5.2.3.

3.4.4 Adhesion.- The primer shall show good adhesion when tested as specified in 4.5.2.4.

3.4.5 Condition in container.- The product shall be capable of being readily broken up with a paddle to a smooth, uniform consistency, and shall not liver, thicken, curdle, gel, nor show any other objectionable properties for at least one year after date of manufacture.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as stipulated in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot.- For purpose of quality assurance provisions a lot shall consist of all primer offered for delivery at one time.

4.3 Sampling.-

4.3.1 Ingredient materials.- A one-pint sample from each lot of the vinyl resin and, when requested by proper authority, a one pint sample from each lot of the other ingredient materials shall be taken by the Government representative for test purposes.

4.3.2 Finished primer.- The procedure for sampling the finished primer for tests shall be in accordance with method 1021 of Standard FED-STD-141.

4.4 Containers.- Sample filled containers (quart or gallon) selected on the basis of four out of each 100 in the lot shall be examined for defects of construction of the container and the closure, for evidence of leakage, for unsatisfactory markings, and all other preparation for delivery requirements. Each sample filled container shall also be weighed to determine the amount of contents. Any container having one or more defects, or under required fill, shall be rejected, and if the number of defective containers exceeds two per each hundred in the lot, the whole lot shall be rejected.

4.5 Test procedures

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4.5.1 Ingredient materials.- The vinyl resin shall be tested to determine compliance with note 2 of Table I. Other ingredient materials submitted shall be tested to determine compliance with the applicable specifications.

4.5.2 Finished primer.-

4.5.2.1 The following tests shall be conducted in accordance with Standard FED-STD-141 to determine conformance with the applicable requirements of this specification.

Tests	FED-STD-141 Method
Pigment (using extraction mixture "C")	4021
Volatiles	4041
Nonvolatile vehicle (calculated by difference)	4053
Water	4081
Coarse particles and skins (using 1:1 toluene-methyl isobutyl ketone)	4091
Weight per gallon	4184
Viscosity	4281
Flash Point (Pensky-Martens Closed Cup)	4293
Fineness of grind	4411
Odor	4401
Red lead	7071

4.5.2.2 Drying time.- Drying time of the finished primer shall be determined by method 4061 of Standard FED-STD-141, except that the specified conditions of temperature and humidity shall apply only for referee tests in case of dispute. Other tests shall be conducted under prevailing laboratory conditions.

4.5.2.3 Compatibility.- Compatibility of the finished primer with thinner shall be determined in accordance with method 4203 of Standard FED-STD-141. Fifty milliliters (ml.) of primer and fifty milliliters of vinyl paint thinner conforming to Specification MIL-T-19588 (1:1 toluene-methyl isobutyl ketone) shall be used. Observations shall be made immediately after mixing and also 30 minutes after mixing.

4.5.2.4 Adhesion.- Apply a coat (0.5 mil dry) of pretreatment primer coating (Spec. MIL-P-15328) to a steel panel, and allow to dry for one hour, then apply the sample vinyl-red lead primer to yield a dry film thickness of approximately 1 mil. After a twenty-four hours dry at 20 degrees Centigrade (68 degrees Fahrenheit), the film shall be subjected to the knife test conforming to Method 6304 of FED-STD-141 to determine whether the primer exhibits good adhesion to the pretreatment coating.

4.5.2.4.1 As an alternative to the knife test, make a cut at least one inch long through the combined paint film with a sharp razor blade, then press a piece of cellulose (Scotch) tape (3/4-inch width) firmly against the paint film over the cut section so that the length of the tape is perpendicular to the cut and so that the cut extends beyond the edges of the tape.

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The tape shall be in contact with the paint film for a distance of at least one inch on each side of the cut and there shall be sufficient excess tape on one side to be held firmly between the thumb and forefinger. Make the test by pulling the tape free in a slow and even manner in a direction at right angles to the panel. Not more than 1/8-inch of the film shall be picked up by the tape.

4.6 Rejection.- Lots shall be accepted or rejected by the Government inspector on the basis of the laboratory test report on the transmitted samples.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- Primer shall be packaged and packed in accordance with level C of Specification TT-P-143. Unless otherwise stipulated, the primer shall be delivered in 1-gallon cans or 5-gallon pails, as stipulated.

5.2 Marking.- Containers shall be marked in accordance with Specification TT-P-143 for military agencies, with the added following data: contract numbers and Federal stock number. Also, marking pertaining to thinning and application instructions shall be as follows:

> "For application instructions see
> Coast Guard Paint and Color Manual
> (CG-263), Article 4-2-3."

5.2.1 To the marking prescribed by Specification TT-P-143 shall be added:

> "PRECAUTIONS.- Normal ventilation is always necessary. When air temperature exceeds 40 degrees Fahrenheit take extra precautions to insure sufficient ventilation for low vapor concentration. Enforce precautions to keep open flames and sparks away from painting and mixing areas."

5.2.2 Unless otherwise stipulated, shipping containers shall be marked with the name of the material, color(s), quantity and size(s) of containers therein, the stock number(s), the name of the contractor, the number of the contract or order, date of manufacture (month, year), and the gross weight.

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6. NOTES

6.1 Intended use.- This specification covers a paint intended for use as an anti-corrosive coating only over a surface previously treated with pretreatment primer (Spec. MIL-P-15328). When used over pretreatment primer, it is suitable for surfaces above and below water. This paint is to be applied by spray or brush, and is intended to serve as a base for a film of either vinyl-alkyd paint or vinyl anti-fouling paint. It is not to be used over conventional paints nor as a substitute for red lead in conventional paint systems.

6.2 Ordering data.- Procurement documents should stipulate the following:

(a) Title, number, and date of this specification.
(b) Size of containers required (see 5.1).
(c) Color required (see 1.2 and 3.4.2).
(d) Federal stock numbers.

6.3 Bid samples.- Unless otherwise stipulated, samples with bids are not required.

6.4 Composition by volume.- For information only, the following composition by volume is included:

Ingredients	Gallons per 100 Gallons
Red Lead	2.9
Vinyl resin	12.3
Tricresyl phosphate	1.5
Methyl isobutyl ketone	43.3
Toluene	40.0

6.5 Vinyl-red lead primer should be purchased by volume, the unit being a U. S. gallon at 15.5 degrees centigrade (60 degrees Fahrenheit).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing Activity:
U. S. Coast Guard

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